UGANDA CHRISTIAN UNIVERSITY

A Centre of Excellence in the Heart of Africa

AI POWERED COMPUTER VISION-BASED FRAMEWORK FOR REAL-TIME ROAD USER CLASSIFICATION AND INTELLIGENT TRAFFIC SIGNAL CONTROL.

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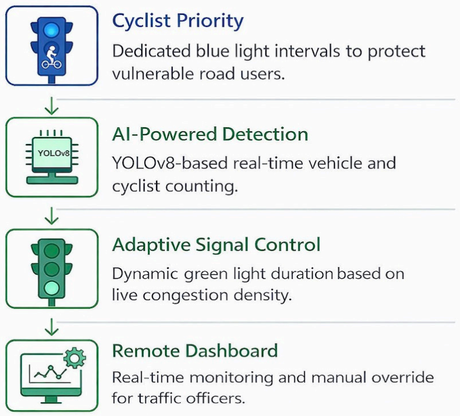
INTRODUCTION

Traffic congestion in Uganda's Greater Kampala Metropolitan Area (GKMA) imposes major economic losses, with daily costs around $1.5 million (≈4.2% of GKMA's daily GDP, per IGC studies) and national impacts estimated at up to 6.7% of GDP annually due to persistent delays and inefficiencies. Many intersections still rely on static traffic signals, leading to prolonged commuter delays and heightened safety risks. This project proposes a computer vision-based intelligent traffic management system that detects road users (vehicles, cyclists), congestion density, and flow patterns in real time to enable adaptive IoT-driven signal control at junctions, aiming to optimize traffic flow, reduce delays, enhance safety, and improve urban mobility and quality of life in Kampala.

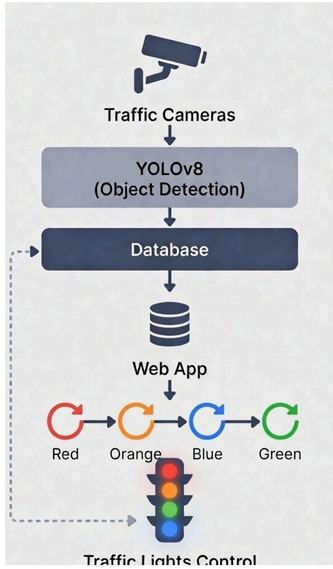
PROBLEM STATEMENT

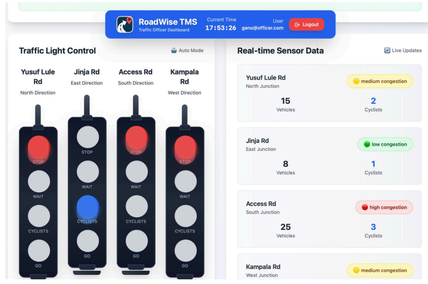
About 60% of intersections in Kampala are severely congested during peak hours, affecting productivity, emergency response, and the safety of vulnerable road users. Most lack intelligent systems, often ignoring cyclists who face high accident risks. There is a critical need for an automated traffic control framework that protects all users and empowers data-driven management.

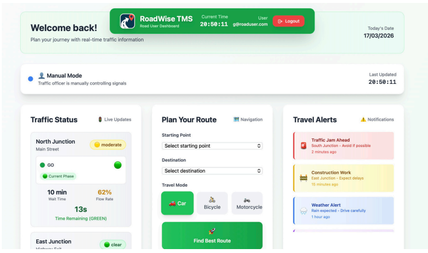
CORE FEATURES



HOW IT WORKS





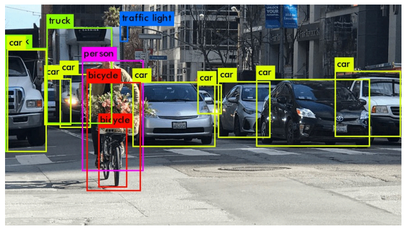


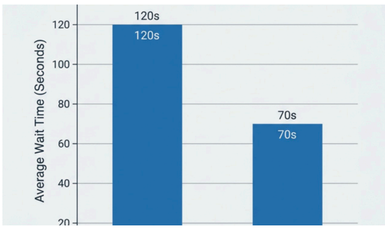
TECHNICAL ARCHITECTURE

- AI & Computer Vision: YOLOv8, OpenCV
- Hardware & IoT: Arduino Uno, NodeMCU, ESP32-CAM
- Web & Cloud: Node.js, Firebase, React
- Languages: C++, Python, JavaScript

IMPACT & FINDINGS

- **Operational Efficiency**: Up to **40% reduction** in average junction wait times through adaptive phase timing.
- **Detection Accuracy:** Achieved **100% vehicle detection** and **85% cyclist detection i**n varied lighting conditions.
- **System Reliability:** Final synchronization protocol achieved **100% state consistency** between web and hardware via Serial ACK feedback.





SDG & NATIONAL DEVELOPMENT PLAN (NDP III) ALIGNMENT

- **SDG 9: Industry, Innovation & Infrastructure** – Building resilient, sustainable road infrastructure through AI.
- **SDG 11: Sustainable Cities & Communities** – Enhancing urban safety and inclusivity for all road users.
- **SDG 3: Health & Well-being** – Reducing road traffic fatalities through prioritized cyclist safety (Target 3.6).
- **Uganda Vision 2040** – Establishing the foundation for a high-tech, modern smart city.
- **Smart City Pillar:** Foundation for KCCA's "High-Tech City" strategic plan for 2020-2025.

REFERENCES

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4. **United Nations**. (2015). *Transforming our world: the 2030*